

ONKAR SARAMBALE

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SUMMARY

Final-year Computer Engineering student with a strong foundation in Python, AI/ML, and data analytics. Experienced in building end-to-end machine learning pipelines, deploying deep learning models (PyTorch, NL2SQL, BERT, ResNet-50), and automating data workflows using AI-powered LLMs. Proven ability to work with large-scale datasets, develop scalable REST APIs, and collaborate in cross-functional teams. Eager to apply AI engineering skills to solve real-world agricultural challenges at JacobAI.

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, C++, HTML, CSS

AI / ML: PyTorch, TensorFlow, Hugging Face, NL2SQL, scikit-learn, BERT, ResNet-50, K-Means, NLP, Deep Learning, Vanna AI 2.0

Data Analytics & Visualization: Pandas, NumPy, SQL, Statistics, Matplotlib, Seaborn, Jupyter

Backend / APIs: Flask, FastAPI, Node.js, Express.js, REST APIs

Cloud & Distributed Systems: Hugging Face Spaces (model deployment), Vercel, REST API hosting, basic AWS familiarity

Tools & Platforms: Git, MongoDB, Docker (basic), Prompt Engineering, Generative AI

PROJECTS

NL2SQL Clinic Chatbot | [LINK](#)

Mar 2026 – Apr 2026

Tech: Python, Vanna AI 2.0, FastAPI, Google Gemini (gemini-2.5-flash), SQLite, Pydantic

- **Built an end-to-end NL2SQL system** converting plain English questions to validated SQL queries using Vanna 2.0 Agent + Google Gemini, served via FastAPI, REST API with Swagger UI
- **Designed a SQL validation layer** blocking dangerous queries (DROP, DELETE, EXEC) before execution;
- pre-seeded DemoAgentMemory with 15 Q&A pairs for improved accuracy
- with 5 tables (200 patients, 15 doctors, 500 appointments) and realistic dummy data with skewed distributions and NULL injection for real-world simulation

Indian Legal Document Classifier | [LINK](#)

Feb 2026 – Mar 2026

Tech: Python, BERT, Flask, React.js, Hugging Face Spaces | [Live Demo](#) | [GitHub](#)

- **Fine-tuned BERT** on a custom Indian legal dataset with 10+ document categories, achieving **92% validation accuracy**, demonstrating applied NLP and transfer learning for domain-specific classification.
- Designed an **end-to-end ML pipeline** covering data preprocessing, tokenization, model training, evaluation, and cloud deployment on Hugging Face Spaces.
- Deployed a full-stack application (Flask + React.js) enabling **real-time inference in under 2 seconds**,
- serving as a scalable AI platform prototype.

NGO Food Donation Platform

Oct 2025 – Jan 2026

Tech: Python, PyTorch, ResNet-50, Flask, React.js, Computer Vision

- Built a **ResNet-50 food recognition model** using PyTorch, achieving **92% accuracy** on the Food-101 dataset (101 classes, 100K+ images) — handling large-scale image data with transfer learning.
- Designed a **real-time image processing pipeline** for automated food recognition and donation tracking, demonstrating applied computer vision for a real-world social use case.
- Integrated a **Generative AI component** to estimate food quantities from images, combining vision models with LLM-based reasoning
- Implemented **location tracking and geo-validation** — fetching donor GPS coordinates and cross-referencing with manually entered addresses to ensure accurate donation pickup routing, demonstrating applied **GIS-adjacent data handling**.

Mood-Based Song Recommendation System

Tech: Python, Pandas, scikit-learn, Flask | GitHub

- Developed a music recommendation system using **K-Means clustering** on **10,000+ Spotify tracks**, extracting insights from audio features (valence, tempo, energy) — statistical analysis at scale
 - **Cleaned and normalized large datasets** using Pandas, applying data transformation and feature engineering best practices for consistent model inference
 - Built and deployed a **real-time Flask REST API** serving predictions with sub-second response time, demonstrating scalable backend design
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INTERNSHIP EXPERIENCE

Data Analytics Intern — VOIS (Vodafone Intelligent Solutions)

Oct 2024 – Nov 2024

- **Automated data analysis workflows** across 5+ business datasets using AI-LLMs, reducing manual analysis time by **~40%** — directly applicable to large-scale agri-data pipelines
- Implemented **AI-powered pipelines** to extract actionable insights and generate structured reports, improving reporting turnaround and decision-making
- Applied **data cleaning and transformation** techniques using Python and Pandas on real-world, messy business datasets — building robust data engineering skills

MERN Stack Web Development Intern — EYGDS & AICTE

Feb 2025 – Mar 2025

- **Built and deployed a full-stack Recipe Sharing Platform** using MongoDB, Express.js, React.js, and Node.js — delivering **8+ REST API endpoints** for auth, CRUD, and real-time data handling
 - Reduced average page load time by **~30%** through optimized component structure and efficient MongoDB queries, demonstrating focus on scalability and system efficiency
 - Collaborated on a **production-grade codebase** following MVC architecture and Git-based version control workflows in a team environment
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EDUCATION

Bachelor of Engineering – Computer Engineering

Nov 2022 – May 2026

KJ College of Engineering and Management Research, Pune | **SGPA: 8.9**

CERTIFICATIONS

- **Machine Learning Specialization** — DeepLearning.AI, Stanford University (Feb 2026)
 - **Intro to Machine Learning** — Kaggle (Jul 2025)
 - **Introduction to Generative AI** — Google (Sep 2024)
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ADDITIONAL INFORMATION

- **Languages:** English, Hindi, Marathi
- **Awards:** Techathon 2.0 by Innovation Foundation Pune (2025) — competed in applied AI problem-solving
- **Interests:** AI research, large-scale data systems, agri-tech innovation, reading, and learning new technologies